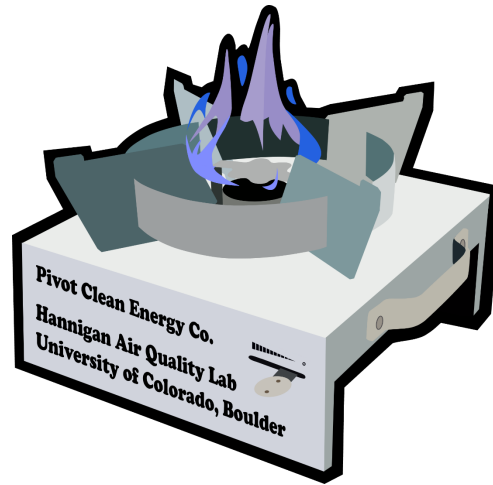
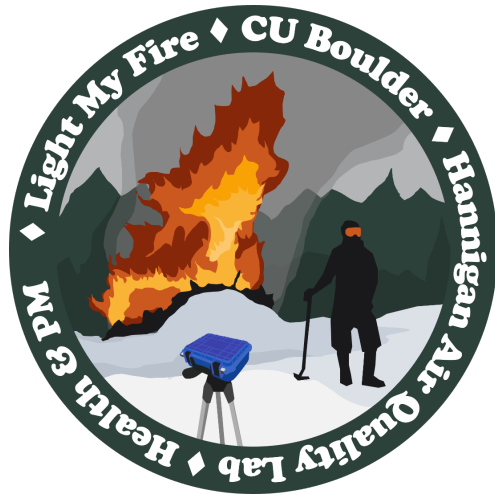




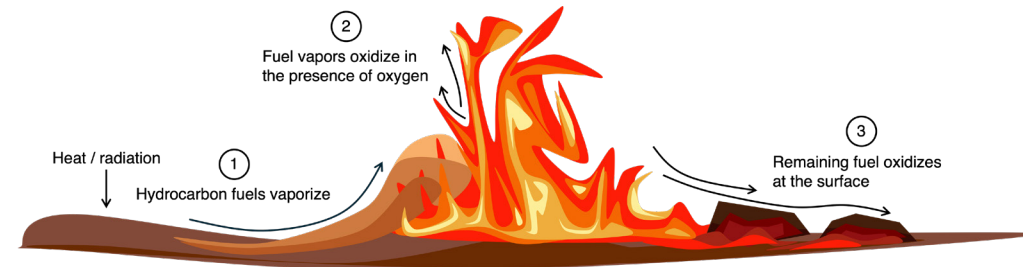
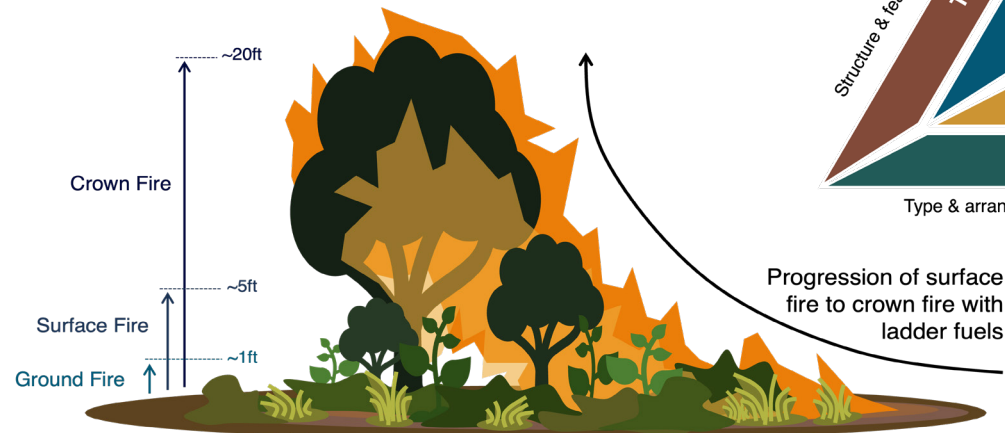
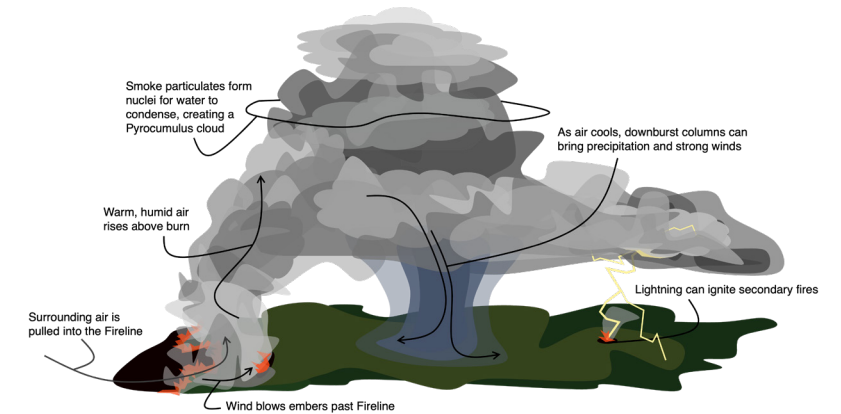
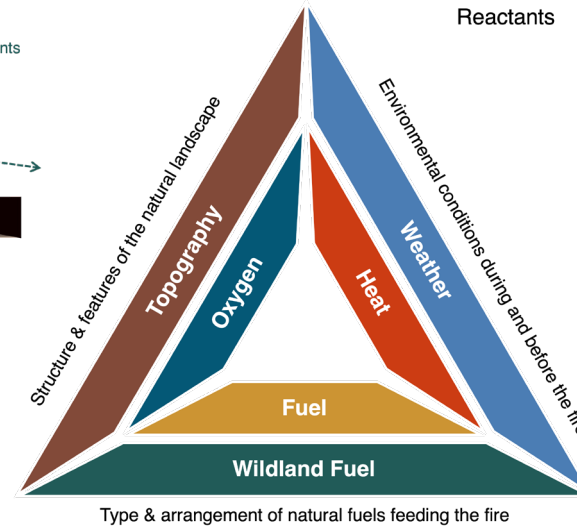
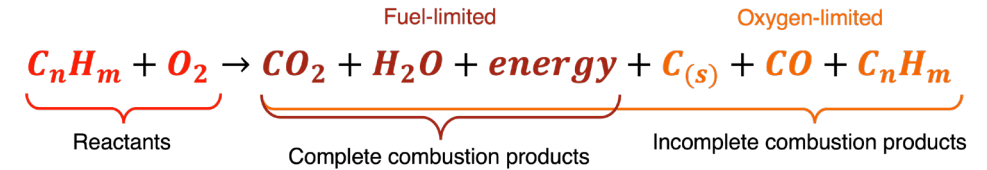
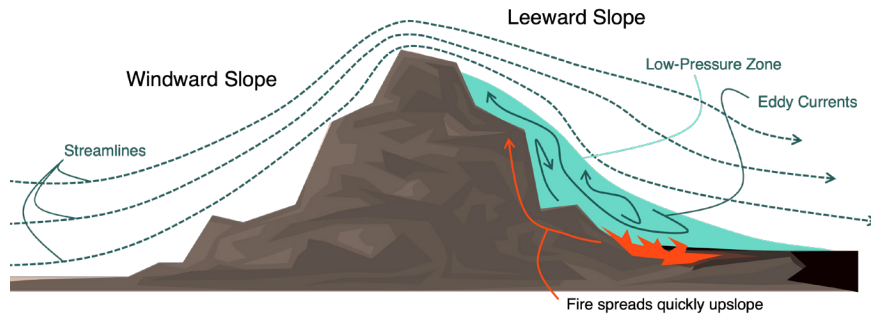
Fire Spread & Firefighting

Central Fire-IQ Slide Deck

Last Updated September 9, 2026 by Chiara Pesce

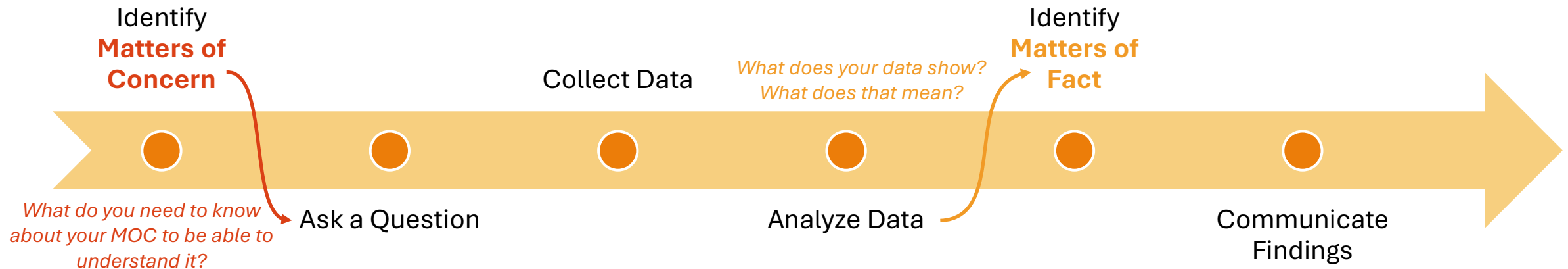


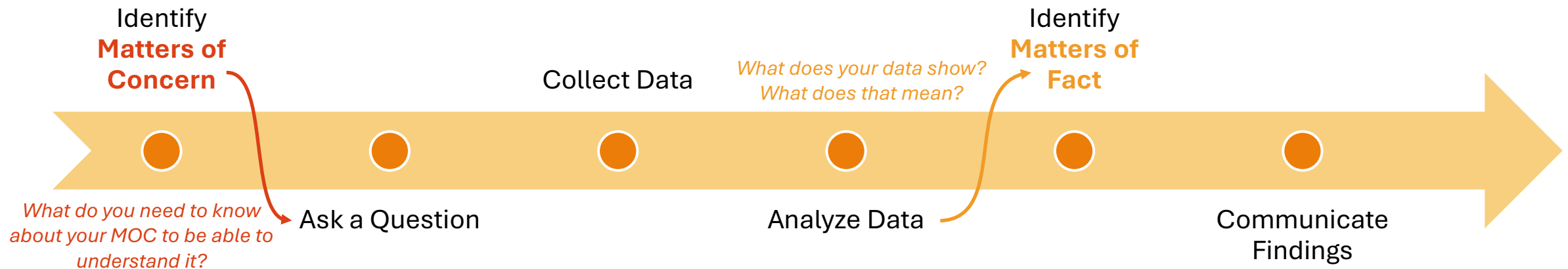
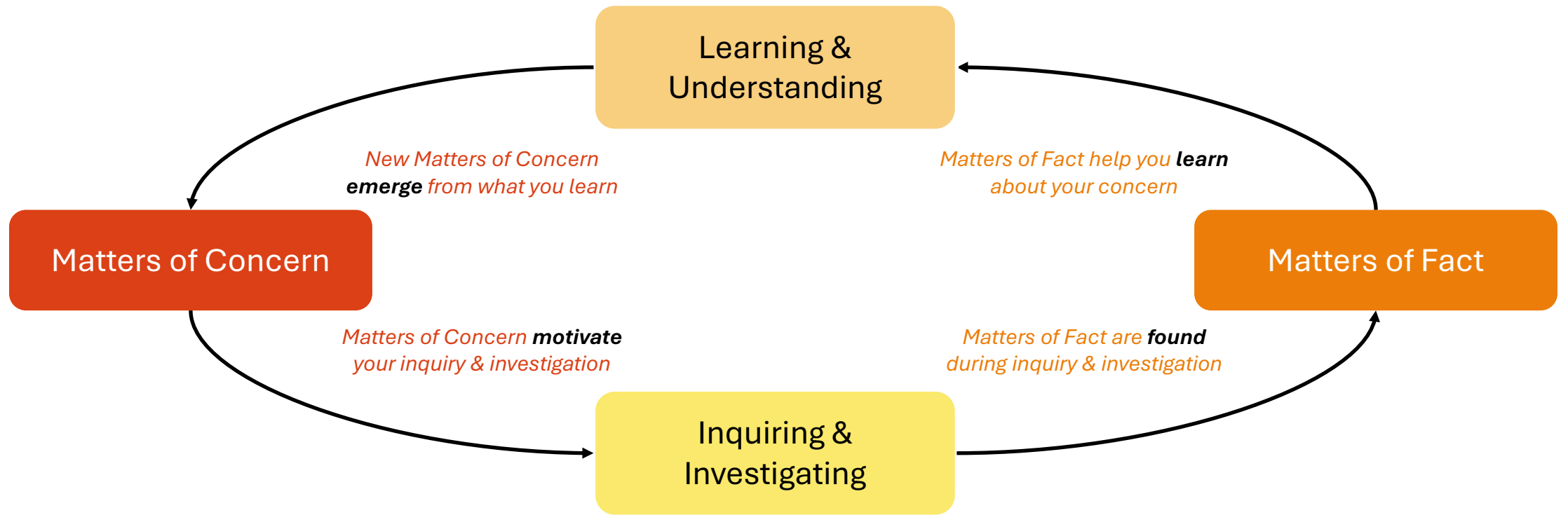
Refresher: Fire Triangle



Matters of Fact and Concern

- **Matters of Concern:** Issues related to a topic that you want to understand
 - Air quality issues that are important to you
- **Matters of Fact:** Conclusions, information & experimental results that define an issue
 - “Knowns” that you use to understand something





Methods of Fire Spread

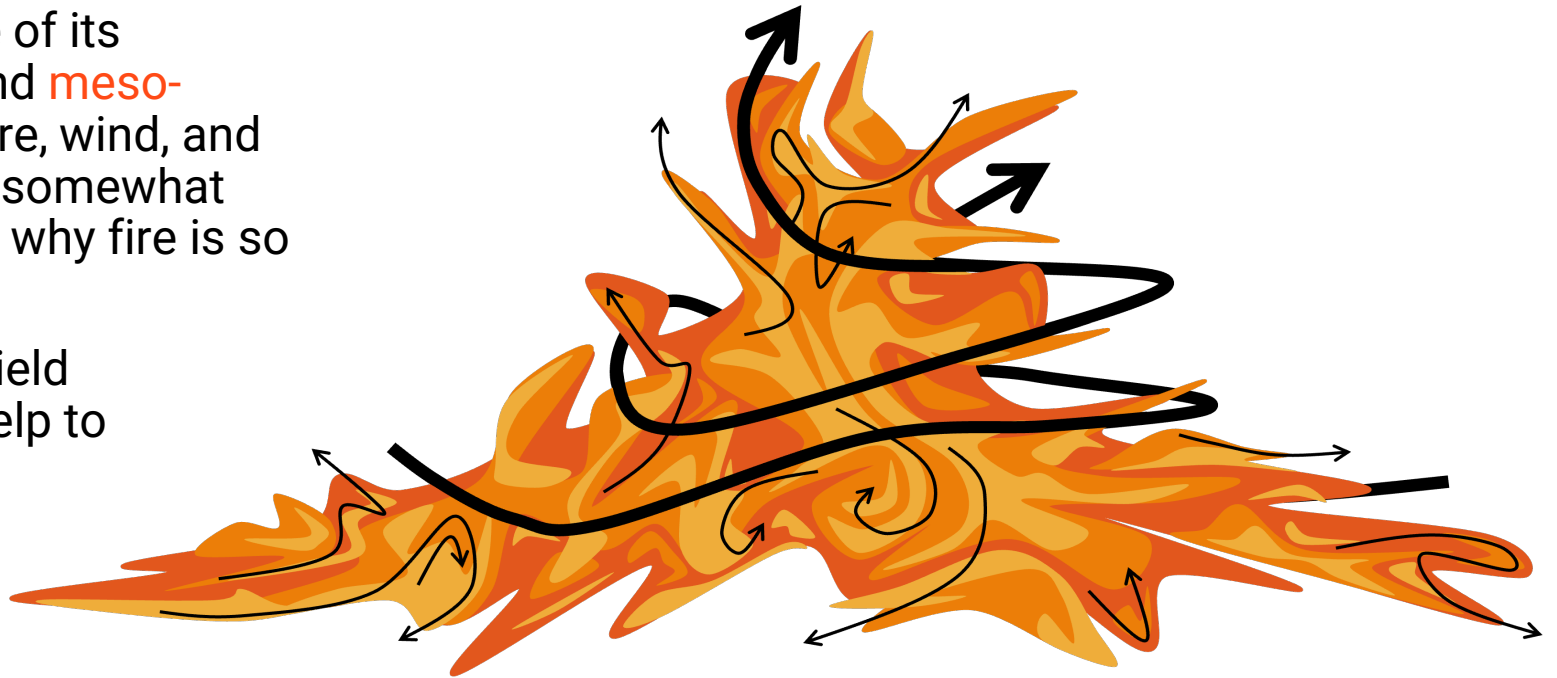


Fire as a Chaotic System

Chaotic system: a complex, dynamic system that is sensitive to initial conditions, inputs and expresses instability

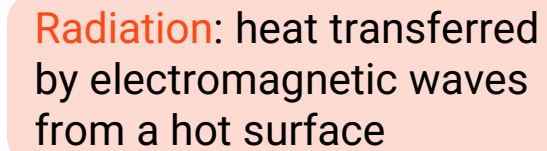
Fire is a chaotic system because of its sensitivity to both **micro-scale** and **meso-scale** changes in fuel, temperature, wind, and weather. This makes fire spread somewhat **unpredictable** and contributes to why fire is so dangerous.

However, research, firefighting, field observations, and science can help to better understand fire spread and behavior



Heat Transfer

There are four main methods of heat transfer through which fire propagates:



Fire **radiates** heat in all directions

Conduction moves heat through surface contact

Convection: heat transferred by the movement of a fluid

Convection moves heat upwards through the air

Direct Contact: heat transferred by direct ignition by flames (basically short-range convection & radiation)

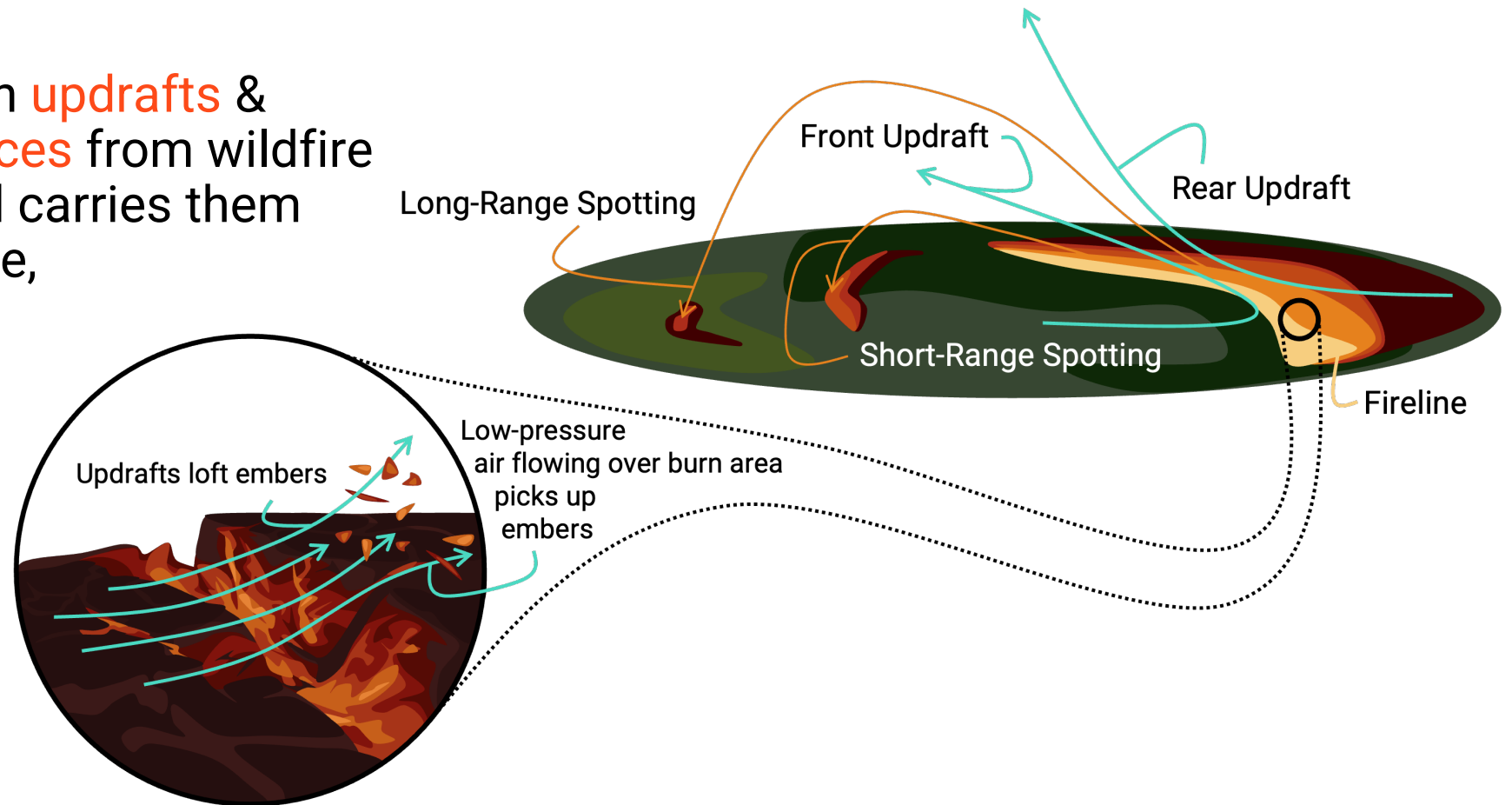
Direct contact with flames increases temperature

Conduction: heat transferred through direct molecular contact

Spotting: a Dangerous Method of Wildfire Spread

Spotting occurs when **updrafts** & **temperature differences** from wildfire heat lifts embers and carries them away from the Fireline, igniting new fires.

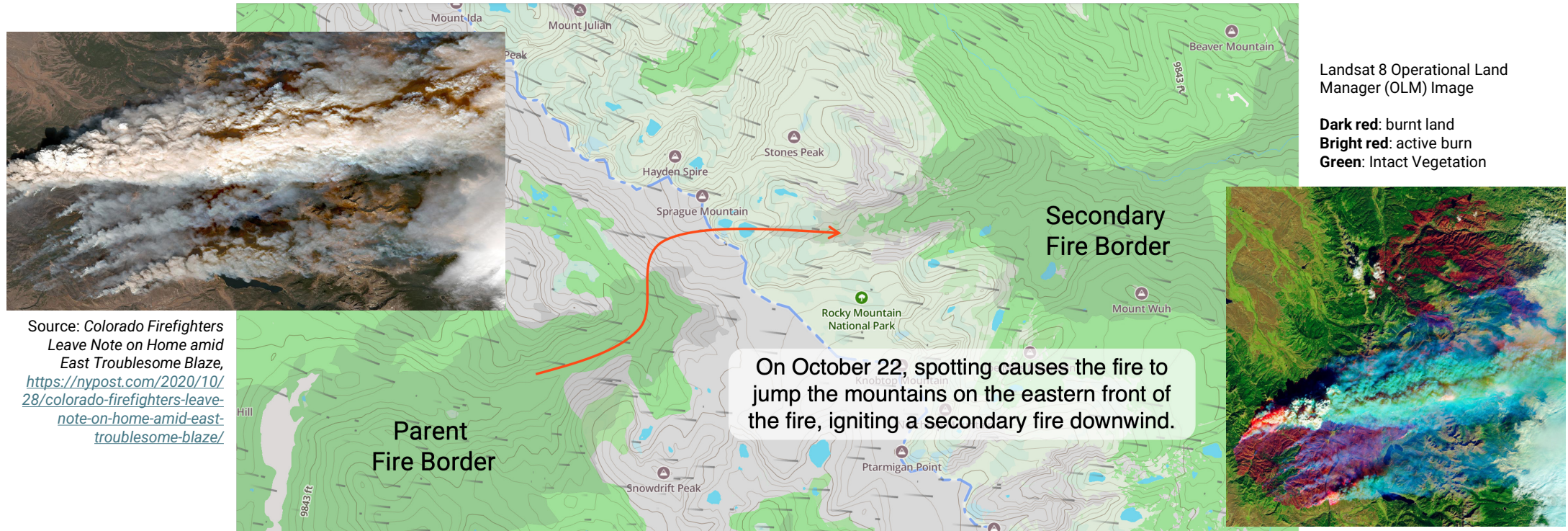
Many structures in the WUI ignite due to spotting.



Spotting: a Dangerous Method of Wildfire Spread

East Troublesome Fire: October 14, 2020

193,812 Acre Burn

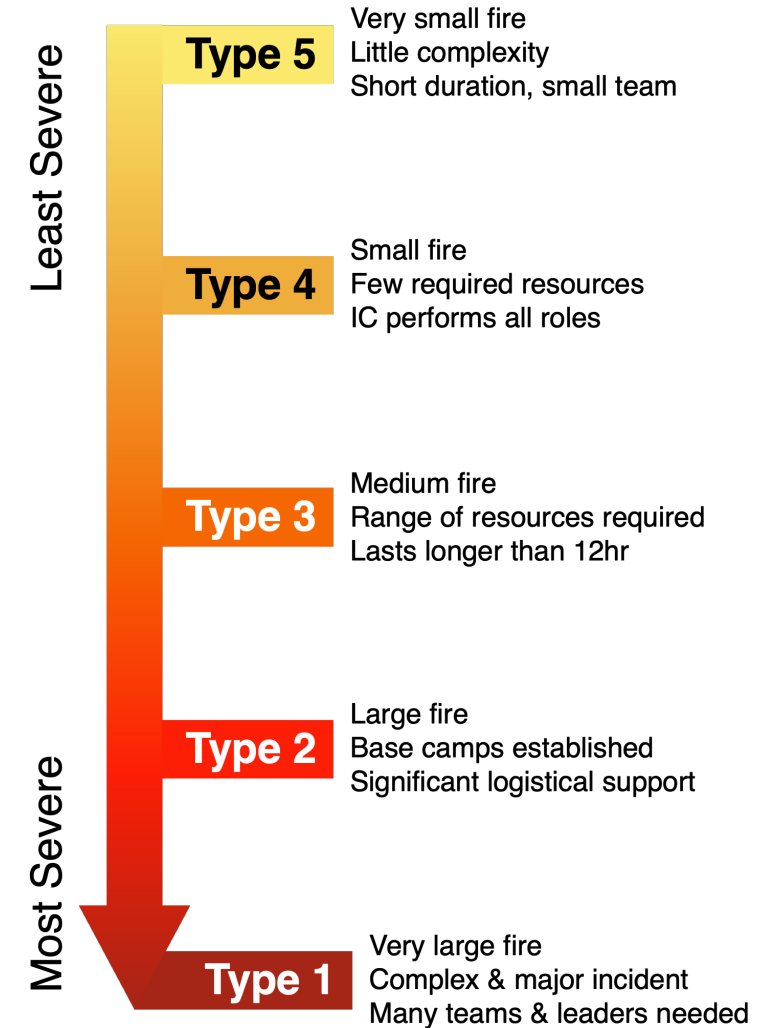


Methods of Firefighting



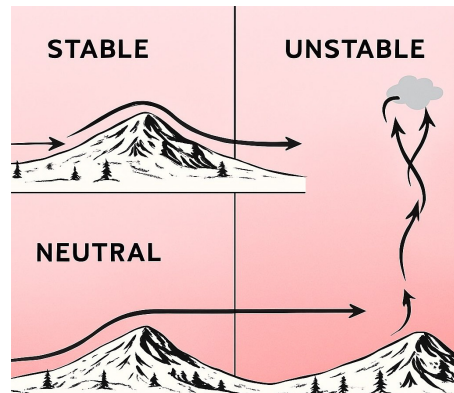
Incident Command System

- The Incident Command System was created in 1970 to **organize** different local, regional or national agencies to respond to wildland fire in an **effective** and **efficient** manner.
- An **Incident Commander** (IC) heads the firefighting effort and is appointed to an incident based on experience, certification and training.
- Incidents are **classified** on a scale from 1-5 based on severity
- Additional specialties (Smokejumpers, aircraft, Hotshots, incident meteorologist etc.) may be recruited based on context

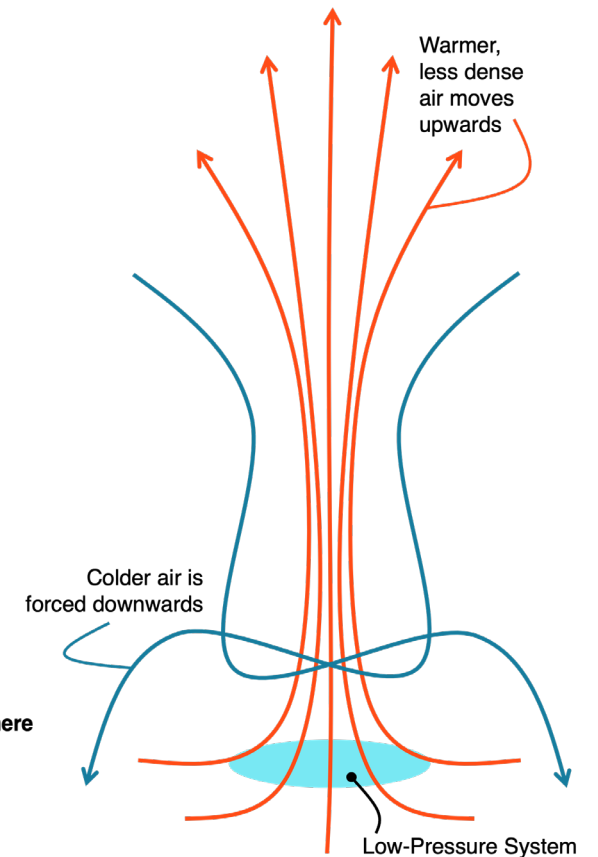
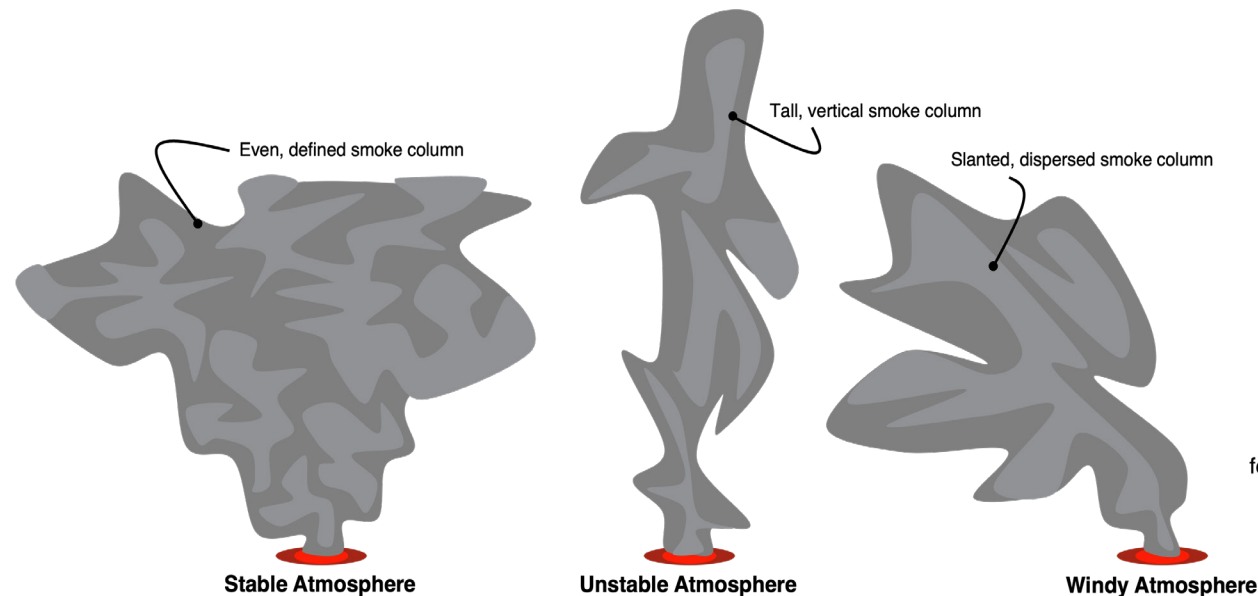


Gathering Information: Atmosphere & Weather

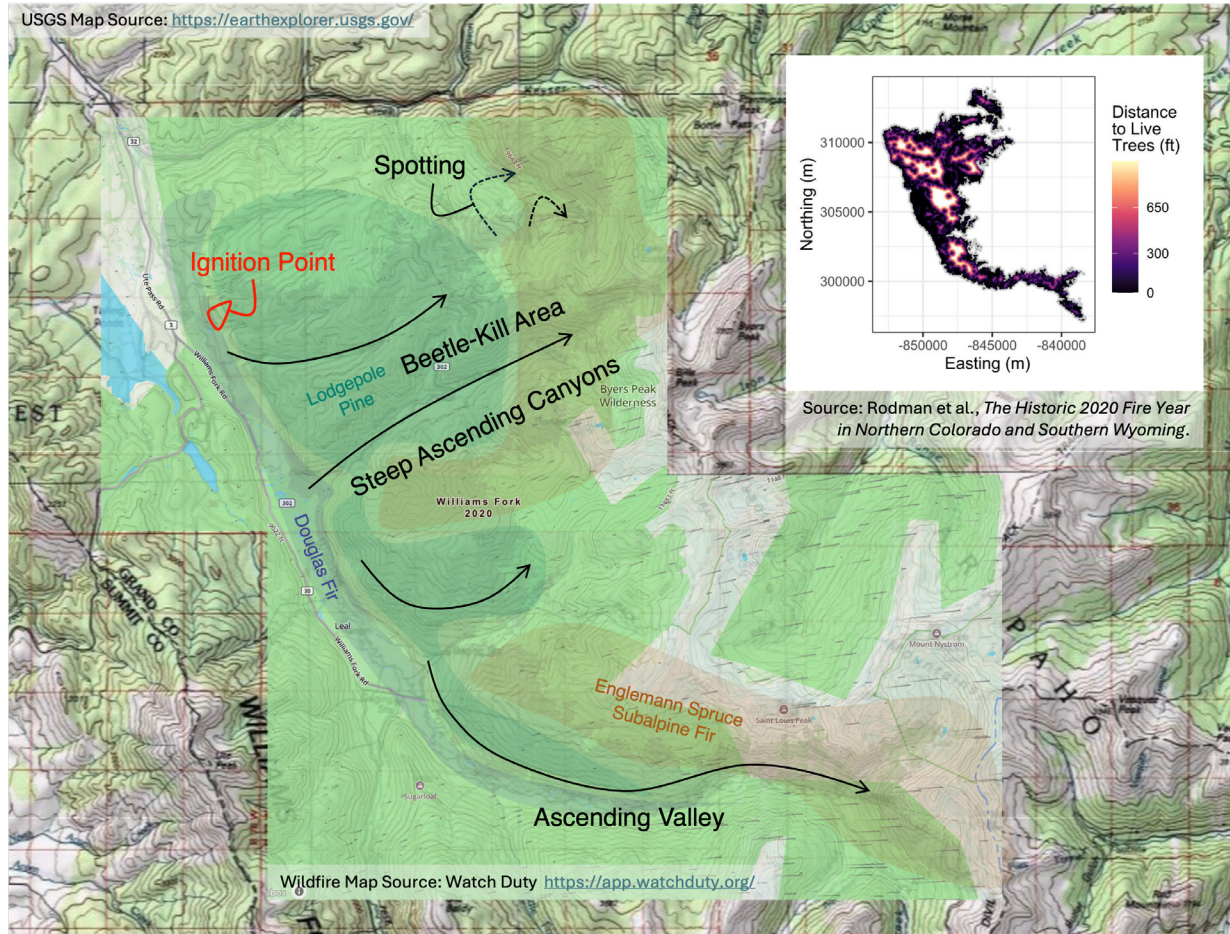
Observations about weather & atmospheric stability can provide key insights into how a wildland fire will behave



Source: "Licchavi Lyceum | Educational Portal." <https://licchavilyceum.com/atmospheric-stability-and-instability-2>



Gathering Information: Fuels & Topography



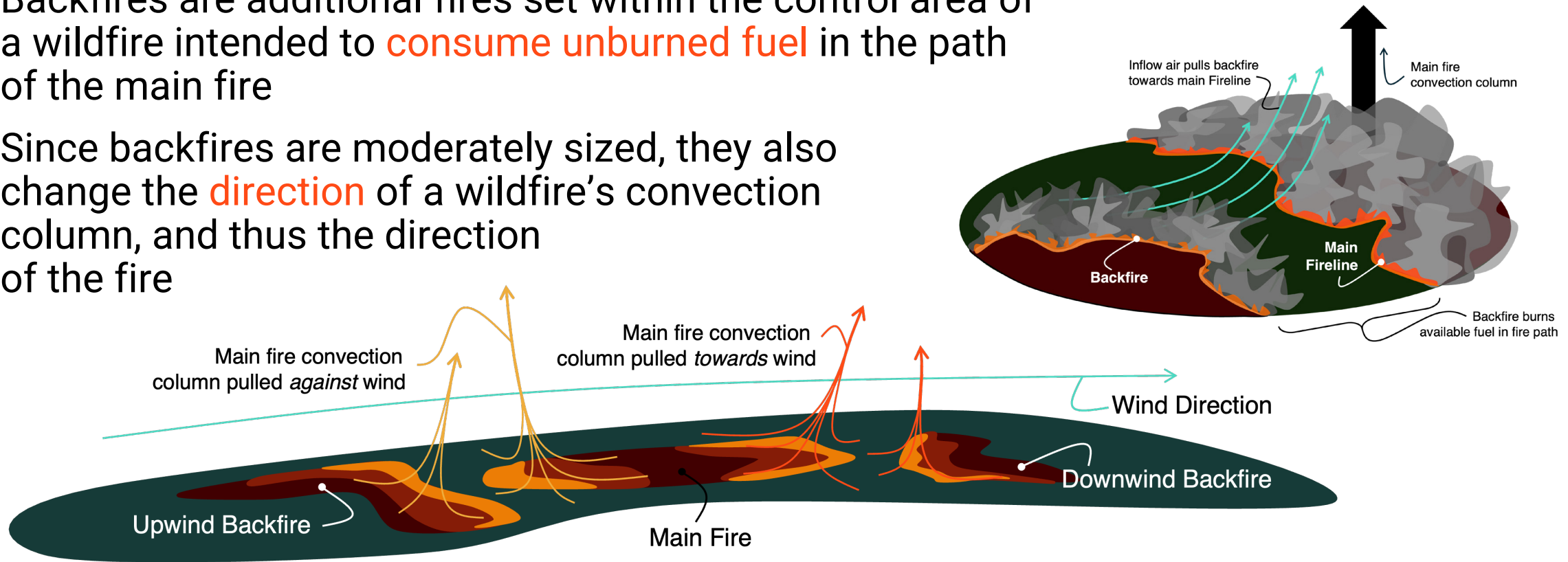
Fire in Malibu, California

Williams Fork Fire August 14th 2020
14,833 Total Acres



Fighting Fire with Fire: Backfires

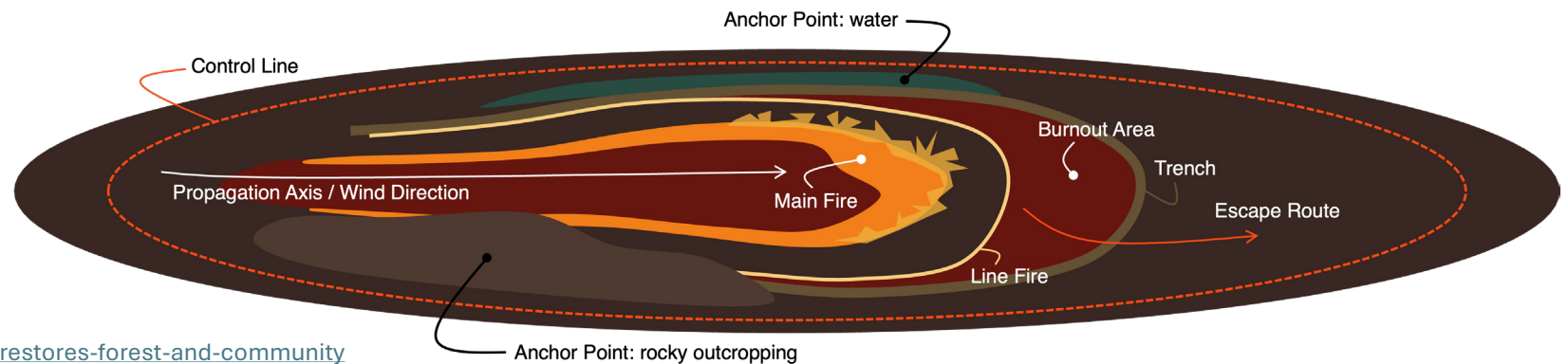
- Backfires are additional fires set within the control area of a wildfire intended to **consume unburned fuel** in the path of the main fire
- Since backfires are moderately sized, they also change the **direction** of a wildfire's convection column, and thus the direction of the fire



Fighting Fire with Fire: Line Fires & Trenching



Source: Bureau of Land Management photo by Joe Bradshaw
<https://www.popsci.com/environment/wildfire-percent-contained/>

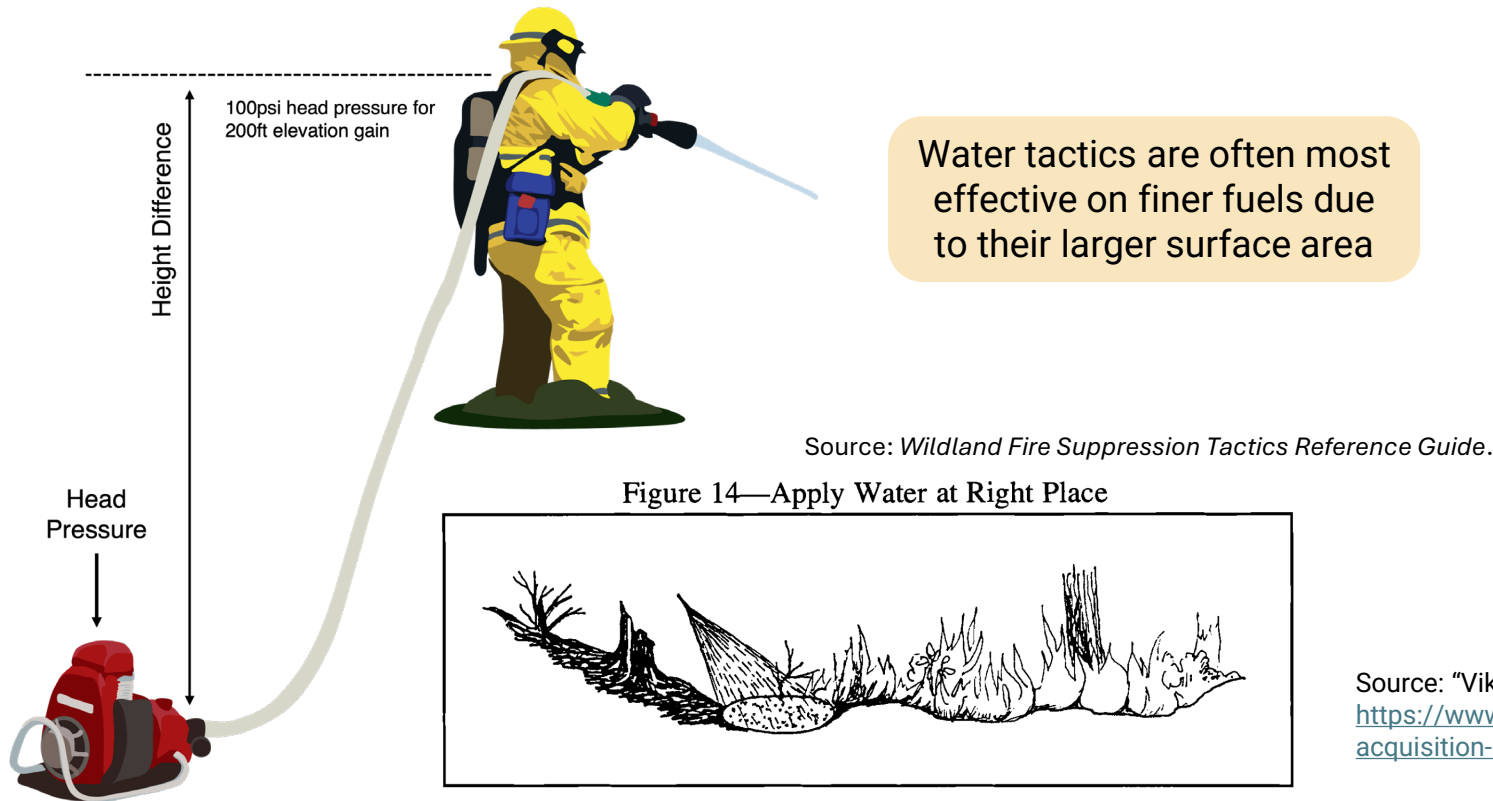


Source: USDA Forest service photo by Matthew Teele
<https://www.fs.usda.gov/about-agency/features/valentine-fire-restores-forest-and-community>



Water-Based Tactics

Water-based firefighting utilizes water's **latent heat of evaporation** to cool the fuels and air around a fire, additionally limiting oxygen supply and increasing the energy needed for combustion.



Source: "Viking Completes Acquisition of Bombardier Amphibious Aircraft Program."
<https://www.aerospacemanufacturinganddesign.com/news/viking-air-completes-acquisition-bombardier-amphibious-100416/>



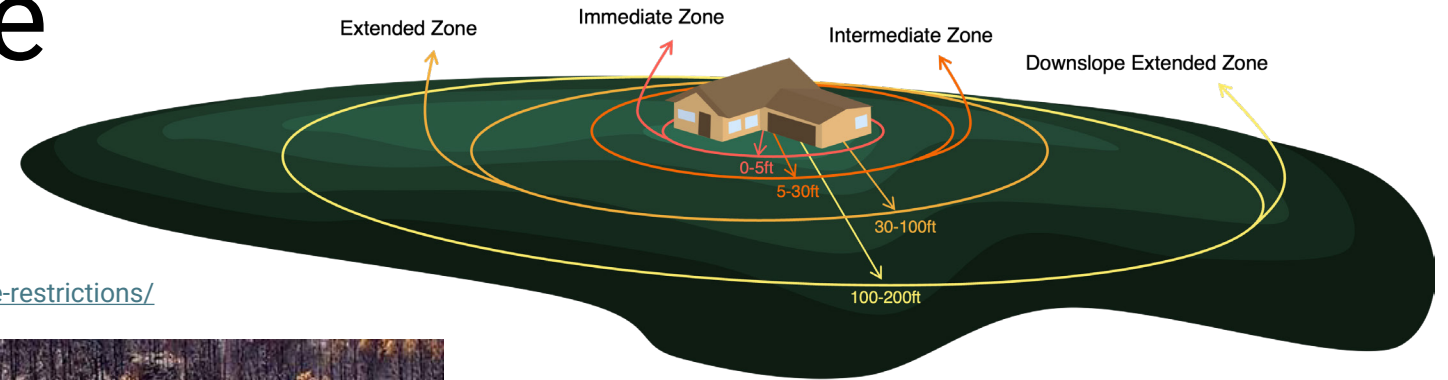
Defensible Space & Preventative Action



Defensible Space

Source: "Current Wildfire Information & Fire Restrictions."

<https://csfs.colostate.edu/wildfire-mitigation/current-wildfire-information-fire-restrictions/>



A surviving home of the
Cold Springs Fire in
Boulder, Colorado 2016



Source: "Will Colorado Communities Adapt to Climate-Driven Wildfire?"

<https://archives.boulderweekly.com/news/will-colorado-communities-adapt-to-climate-driven-wildfire/>



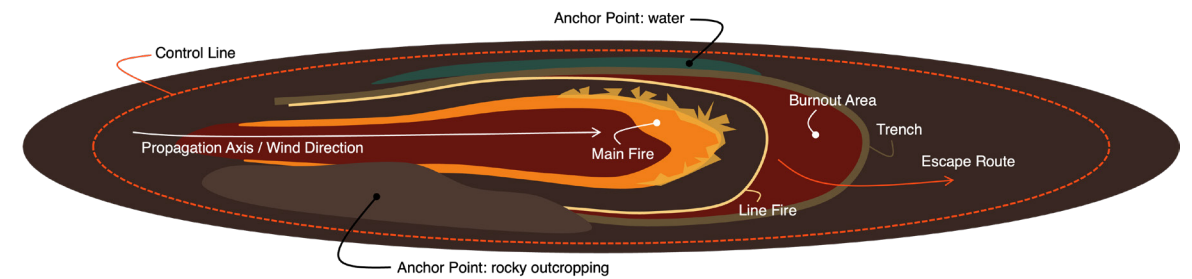
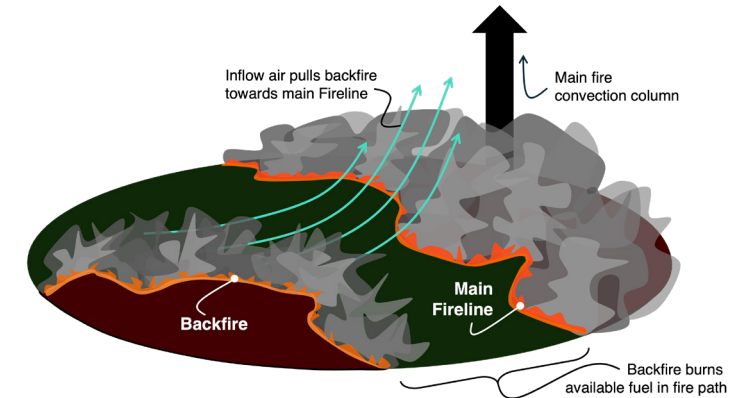
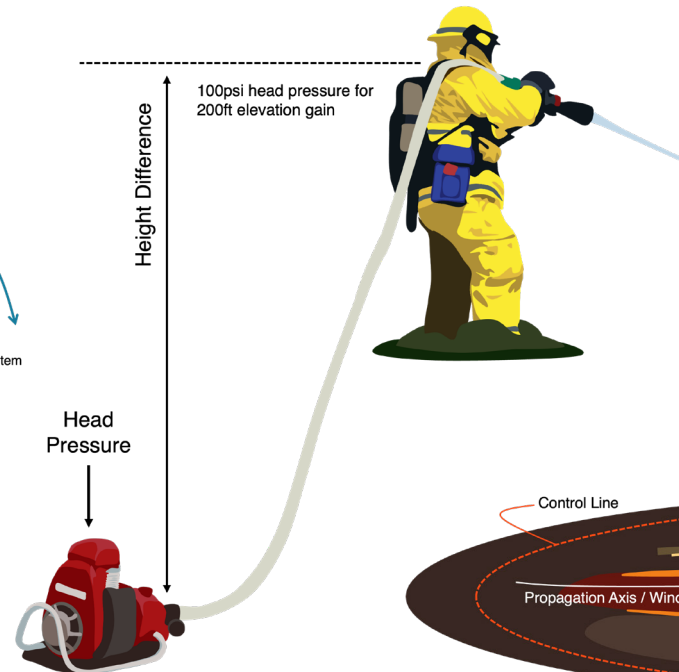
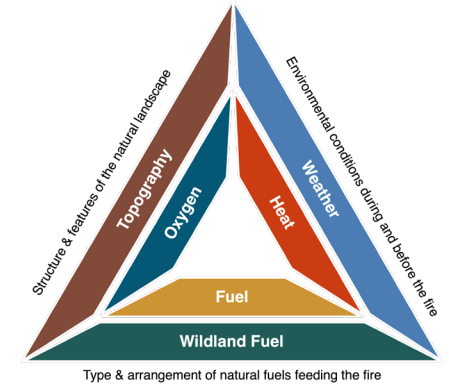
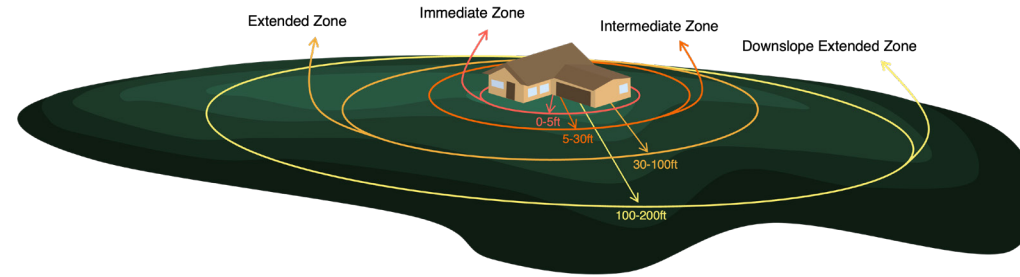
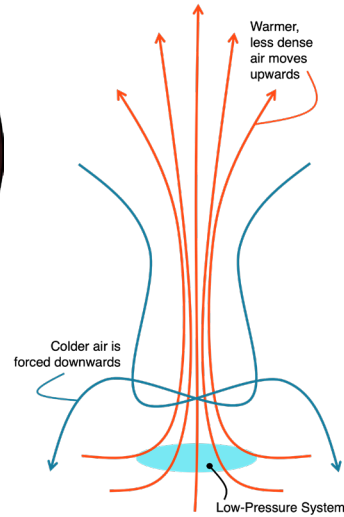
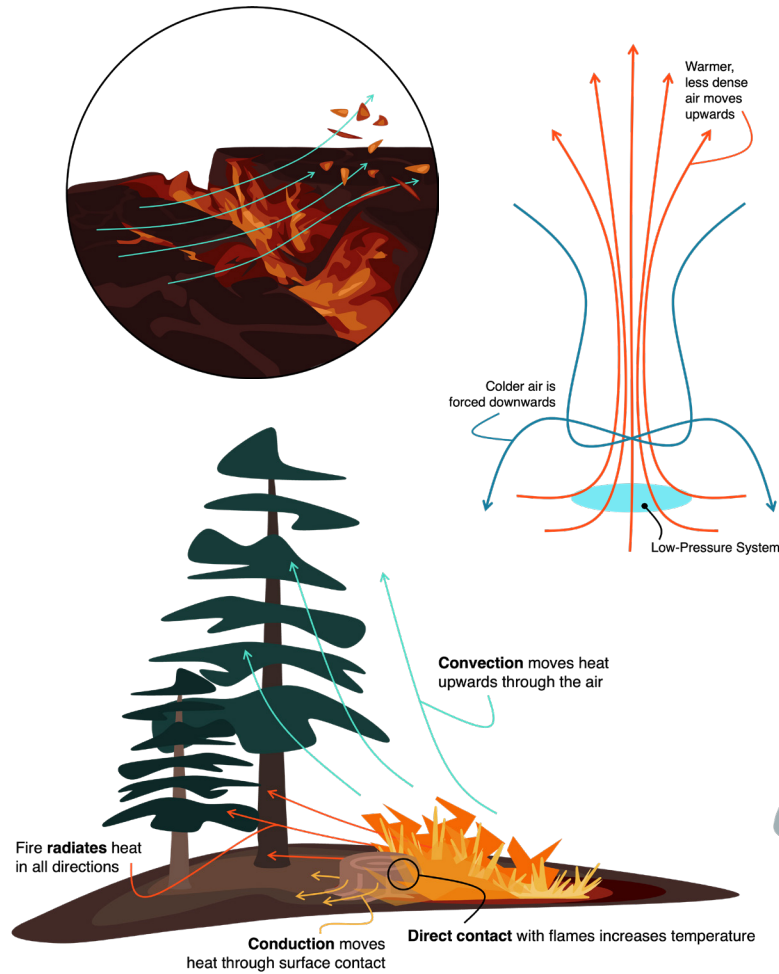
Defensible Space & Preventative Action



Wildfire Mitigation & Spread Matchstick Board Activity

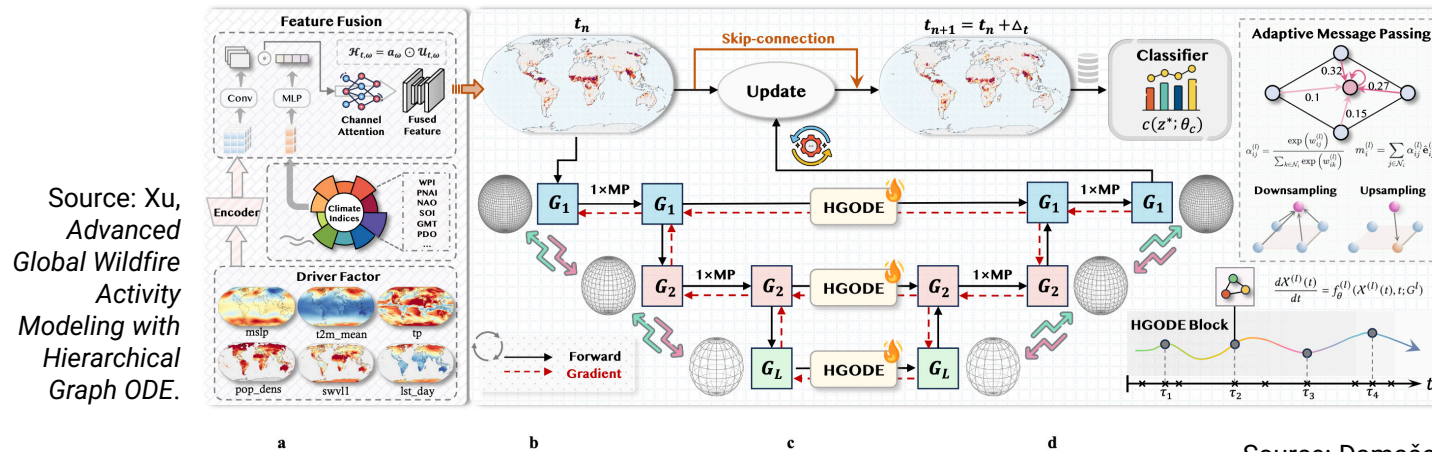


Reminders



Modeling Wildfire

Models can be used to **better understand** or **predict** physical phenomena like wildfire. While helpful, models are often **oversimplified** when compared to reality.



Source: Damaševičius and Maskeliūnas, "Fractional-Order Forest Wildfire Modeling for Predictive Analysis and Risk Mitigation."

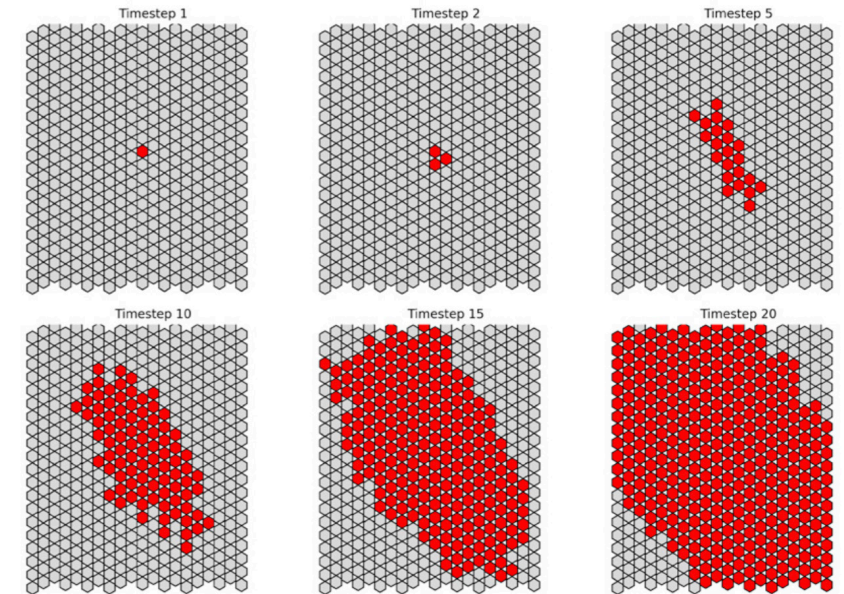
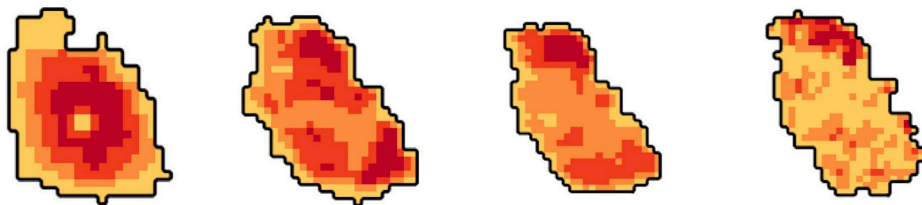


Fig. 3. Fire spread dynamics.

Wildfire spread predictions

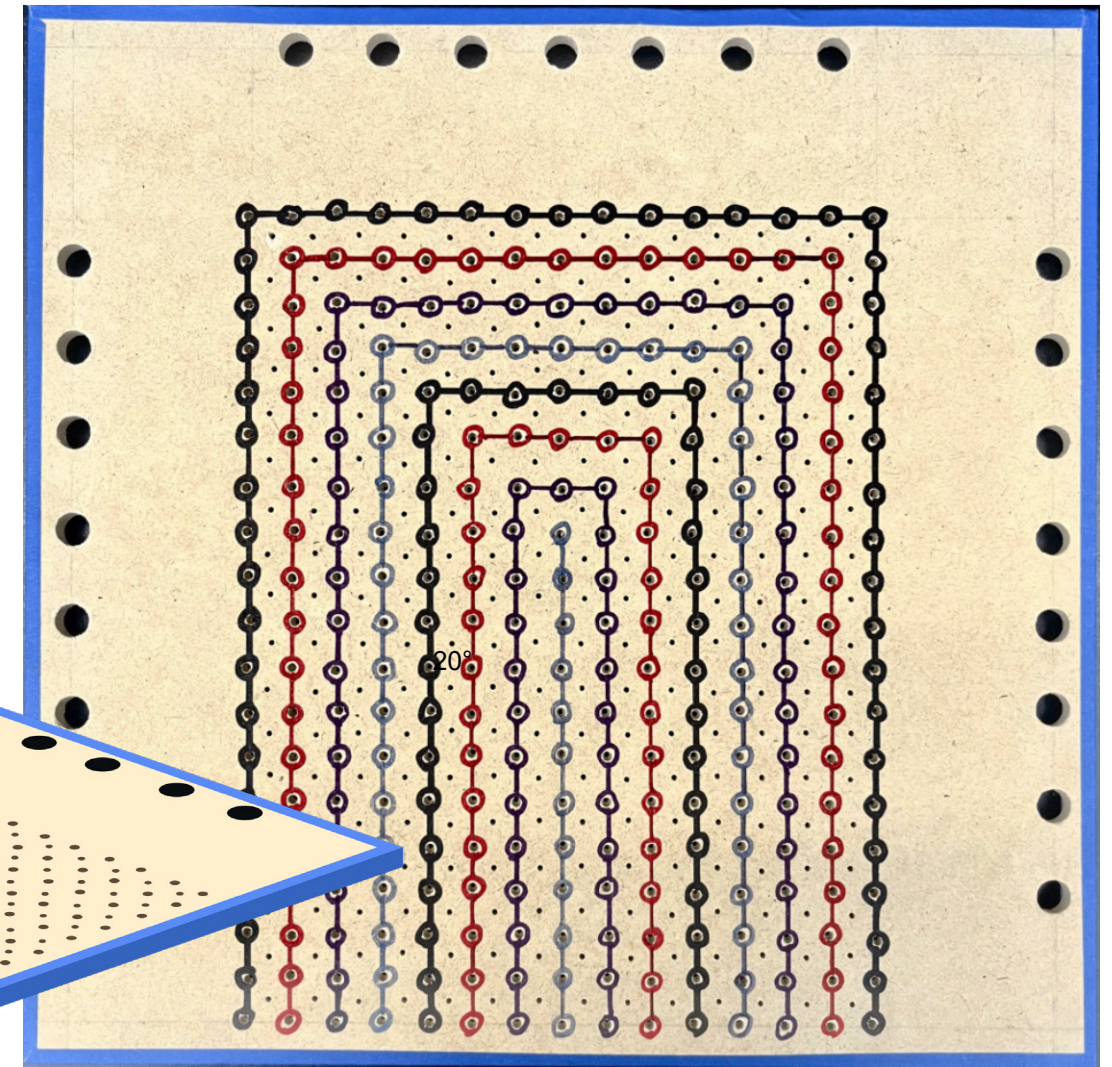
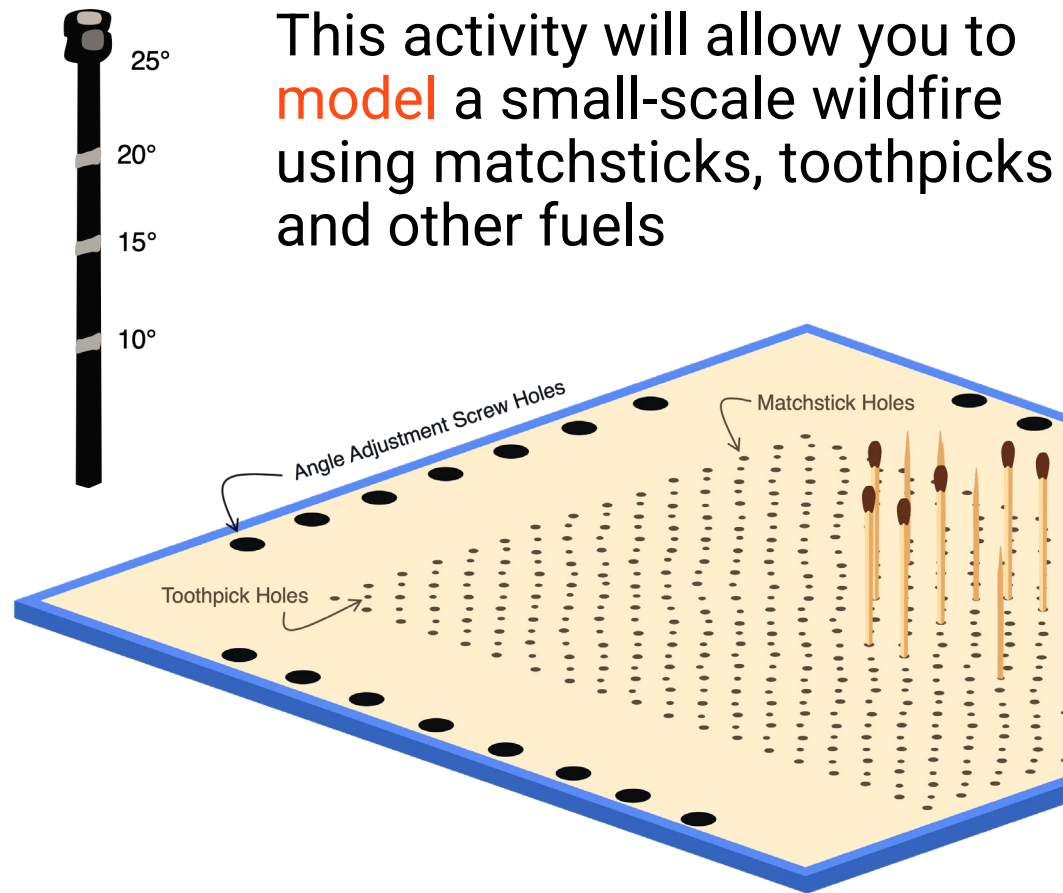


Recall	0.797	0.957	0.946	0.976
Precision	0.895	0.850	0.897	0.94
F1 score	0.843	0.9	0.921	0.958

Source: Nematshahi et al., "High-Fidelity CNN-Based Surrogate Modeling for Wildfire Spread Forecasting."



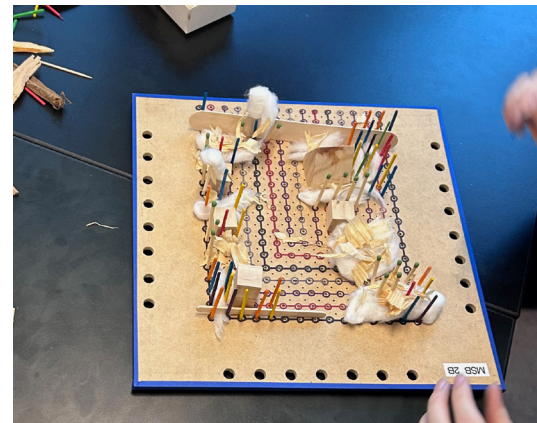
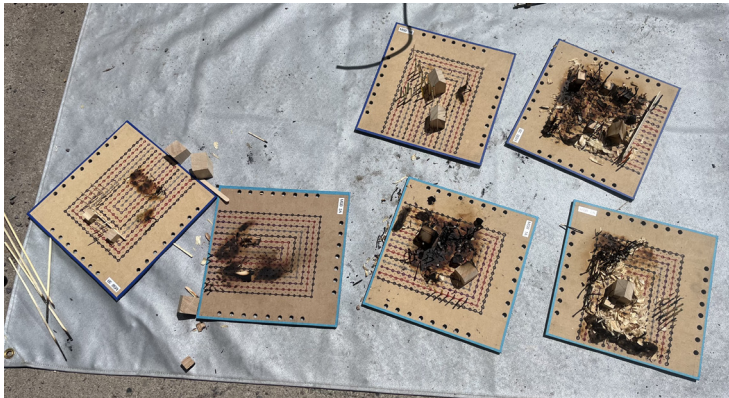
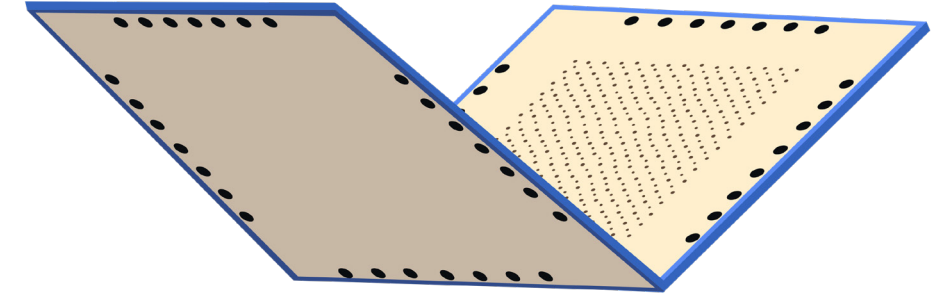
Matchstick Board



Matchstick Board Activity

Available Fuels:

- Toothpicks
- Matches
- Small houses / blocks
- Cotton
- Wood shavings
- Anything else you'd like to add!



Bonus: What could each of these fuels model?



Matchstick Board Activity

Procedure:

1. Select an event / situation or concept to model on the board
2. Choose materials that best fit your scenario
3. Build your scenario! (10-15 minutes)
4. Make predictions about fire behavior on your board
5. Ignite!

Hints & Suggestions

- Boards must be relatively dense to allow fire to propagate
- Use wood shavings or other small materials to simulate ground cover
- What area(s) of the board would benefit from preventative action / defensible space?



If you want to learn more, we recommend starting with these excellent sources!

Wildland Fire Suppression Tactics Reference Guide. n.d.

“Chapter 4 Moisture and Atmospheric Stability.” Accessed August 28, 2026.

<https://www.richhoffmanclass.com/chapter4.html>.

“Defensible Space for Wildfires - Protecting Your Home.” Accessed August 30, 2026.

<https://www.ecologyartisans.com/blog/defensible-space-wildfires>.

East Troublesome Fire – Colorado’s Second Largest Fire on Record – Spreads to the Rockies. October 28, 2020.

<https://scitechdaily.com/east-troublesome-fire-colorados-second-largest-fire-on-record-spreads-to-the-rockies/>.

“Colorado Firecamp, Wildland Fire Suppression Tactics Reference Guide.” Accessed September 5, 2026.

<https://www.coloradofirecamp.com/suppression-tactics/fireline-construction.html>.

“Current Wildfire Information & Fire Restrictions.” *Colorado State Forest Service*, n.d. Accessed September 5, 2026.

<https://csfs.colostate.edu/wildfire-mitigation/current-wildfire-information-fire-restrictions/>.

Damaševičius, Robertas, and Rytis Maskeliūnas. “Fractional-Order Forest Wildfire Modeling for Predictive Analysis and Risk Mitigation.” *Procedia Computer Science* 280 (2026): 15–22. <https://doi.org/10.1016/j.procs.2026.04.004>.

“GridFire.” Accessed September 9, 2026. <https://gridfire.ucr.edu/index.html>.

